



CSE-4508

Relational Database Management System

Lab 6

Lab Group: **1A**

Thursday, 1st January, 2026

Part A: JOIN Operations

Initial Table Setup

EMP_ID	EMP_NAME	DEPT_ID	SALARY	MANAGER_ID
101	John Smith	10	50000	NULL
102	Sarah Johnson	20	60000	101
103	Mike Wilson	10	55000	101
104	Emily Brown	30	65000	101
105	David Lee	NULL	45000	102

DEPT_ID	DEPT_NAME	LOCATION
10	IT	New York
20	HR	Boston
30	Finance	Chicago
40	Marketing	Seattle

Input Table Description

EMPLOYEES Table:

- **EMP_ID**: Unique identifier for each employee.
- **EMP_NAME**: Name of the employee.
- **DEPT_ID**: Department identifier where the employee works. NULL indicates no department assignment.
- **SALARY**: Employee's annual salary.
- **MANAGER_ID**: Employee ID of the direct manager. NULL indicates top-level employee.

DEPARTMENTS Table:

- **DEPT_ID**: Unique identifier for each department.
- **DEPT_NAME**: Name of the department.
- **LOCATION**: City where the department is located.

Task 1

Objective

Display employee names along with their department names for employees who are assigned to a department.

Expected Output

EMP_NAME	DEPT_NAME
John Smith	IT
Sarah Johnson	HR
Mike Wilson	IT
Emily Brown	Finance

Task 2

Objective

Display all employees with their department names. Include employees who are not assigned to any department.

Expected Output

EMP_NAME	DEPT_NAME
John Smith	IT
Sarah Johnson	HR
Mike Wilson	IT
Emily Brown	Finance
David Lee	NULL

Task 3

Objective

Display all departments with their employees. Include departments that have no employees assigned.

Expected Output

EMP_NAME	DEPT_NAME
John Smith	IT
Mike Wilson	IT
Sarah Johnson	HR
Emily Brown	Finance
NULL	Marketing

Task 4

Objective

Display all employees and all departments, showing matches where they exist and preserving unmatched records from both tables.

Expected Output

EMP_NAME	DEPT_NAME
John Smith	IT
Mike Wilson	IT
Sarah Johnson	HR
Emily Brown	Finance
David Lee	NULL
NULL	Marketing

Task 5

Objective

Create all possible combinations of employees and departments (Cartesian Product).

Expected Output (First 10 rows shown)

EMP_NAME	DEPT_NAME
John Smith	IT
John Smith	HR
John Smith	Finance
John Smith	Marketing
Sarah Johnson	IT
Sarah Johnson	HR
Sarah Johnson	Finance
Sarah Johnson	Marketing
Mike Wilson	IT
Mike Wilson	HR
... (20 rows total)	

Explanation

CROSS JOIN returns the Cartesian product of both tables, creating every possible combination. In this result:

- With 5 employees and 4 departments, it produces $5 \times 4 = 20$ total rows.
- Each employee is paired with every department, regardless of their actual department assignment.
- This type of join has limited practical use but can be helpful for generating test data, creating all possible scenarios for analysis, or when you need to compare every item from one set with every item from another set.

Part B: String Functions

Initial Table Setup

CUST_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE
1	john	SMITH	john.smith@email.com	555-1234
2	SARAH	johnson	(space)sarah@email.com(space)	555-5678
3	mike	WILSON	mike.wilson@email.com	555-9012

Input Table Description

CUSTOMERS Table:

- **CUST_ID**: Unique identifier for each customer.
- **FIRST_NAME**: Customer's first name (inconsistent capitalization).
- **LAST_NAME**: Customer's last name (inconsistent capitalization).
- **EMAIL**: Customer's email address (may contain leading/trailing spaces).
- **PHONE**: Customer's phone number.

The notation (space) indicates the presence of leading and/or trailing spaces in the data for CUST_ID = 2. If representing spaces causes formatting issues, an underscore (_) may be used instead.

Note

You should use the following built-in functions:

COUNT
UPPER
CONCANT
INSTR
SUBSTR
LOWER
LENGTH
L/R PAD
INITCAP
L/R TRIM

Task 6

Objective

Combine FIRST_NAME and LAST_NAME to create a FULL_NAME column with a space separator.

Expected Output

FULL_NAME
john SMITH
SARAH johnson
mike WILSON

Task 7

Objective

Properly capitalize the first and last names with the first letter uppercase and remaining letters lowercase.

Expected Output

FIRST_NAME	LAST_NAME
John	Smith
Sarah	Johnson
Mike	Wilson

Task 8

Objective

Convert first names to uppercase and email addresses to lowercase for standardization.

Expected Output

FIRST_NAME	EMAIL
JOHN	john.smith@email.com
SARAH	(space)sarah@email.com(space)
MIKE	mike.wilson@email.com

Task 9

Objective

Calculate the total number of characters in each email address.

Expected Output

EMAIL	EMAIL_LENGTH
john.smith@email.com	20
(space)sarah@email.com(space)	17
mike.wilson@email.com	21

Task 10

Objective

Format phone numbers to a total width of 15 characters using padding with specified characters.

Expected Output

PHONE	LEFT_PADDED	RIGHT_PADDED
555-1234	*****555-1234	555-1234——-
555-5678	*****555-5678	555-5678——-
555-9012	*****555-9012	555-9012——-

Task 11

Objective

Extract the first 4 characters from each email and extract the domain name (portion after the @ symbol).

Expected Output

EMAIL	FIRST_4_CHARS	DOMAIN
john.smith@email.com	john	email.com
(space)sarah@email.com(space)	(spa	email.com(space)
mike.wilson@email.com	mike	email.com